

SIH26169 · coarse pointing, acquisition & tracking for mobile FSOC terminals · team Regnum Carya

software-in-the-loop simulation ground-truth metrics identical seeds for every pipeline

1 · Executive summary

87 %

runs acquired ≤ 2 s
 baselines —

1.29 s

median acquisition time
baselines —

6/10

scenarios meeting every PS limit
 baselines —

43

slowest closed-loop FPS (PS ≥ 20)
10 scenarios

- Slowest closed loop (simulation + rendering + tracking) ran at **43 FPS**, above the 20 FPS requirement in every scenario.
- **Not met:** J fast (Err); K occlusion (Err); N worst (Acq, Err, Loss); O distractors (Acq). See limitations.

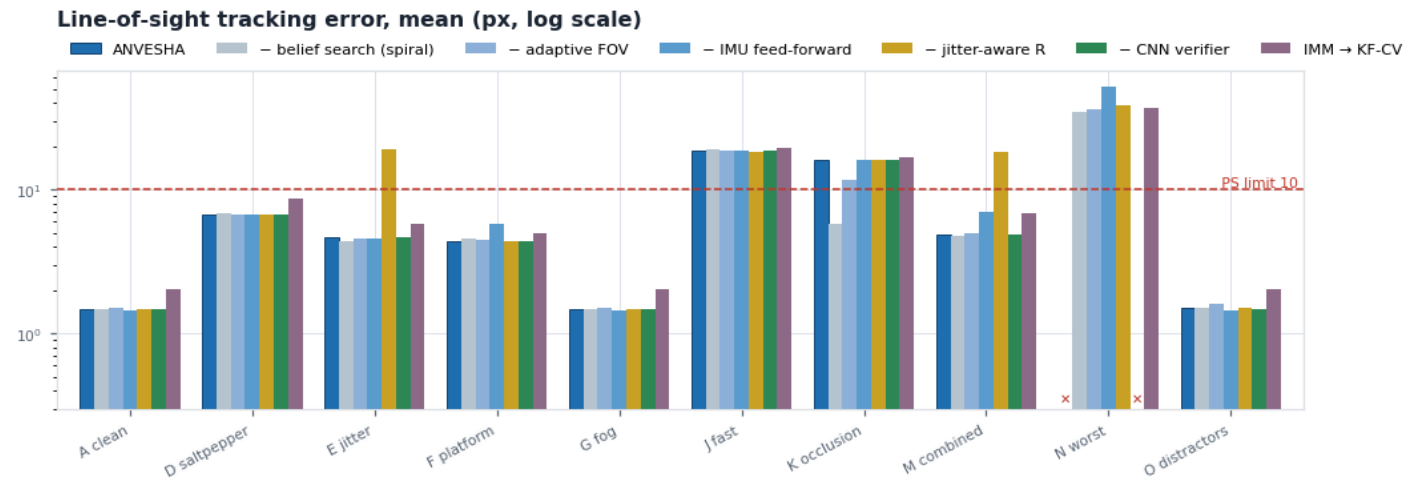
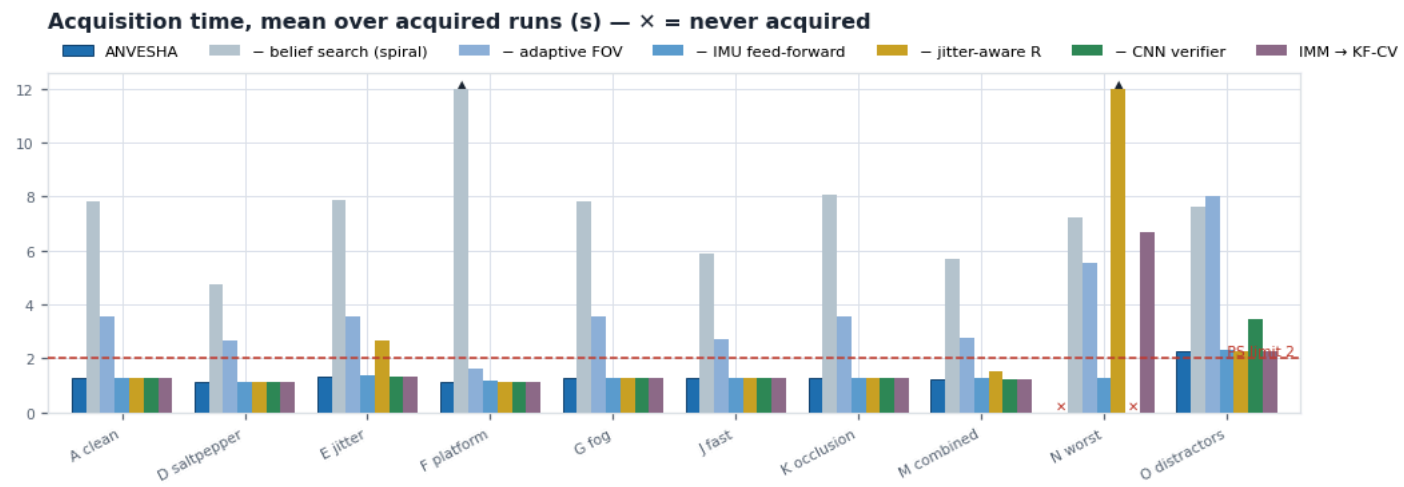
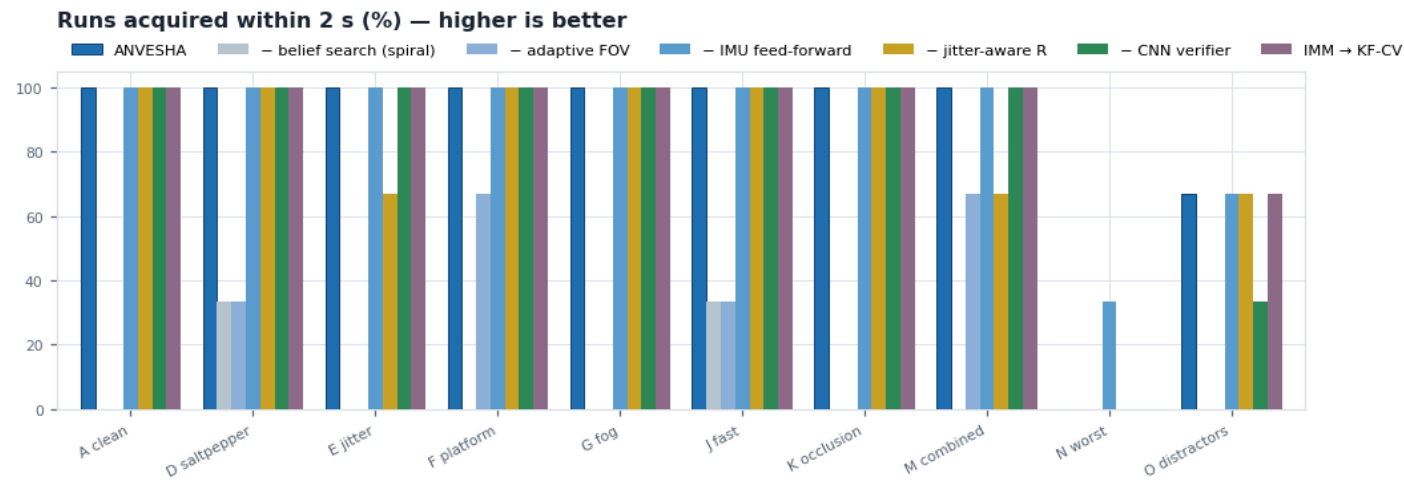
2 · SIH26169 compliance matrix

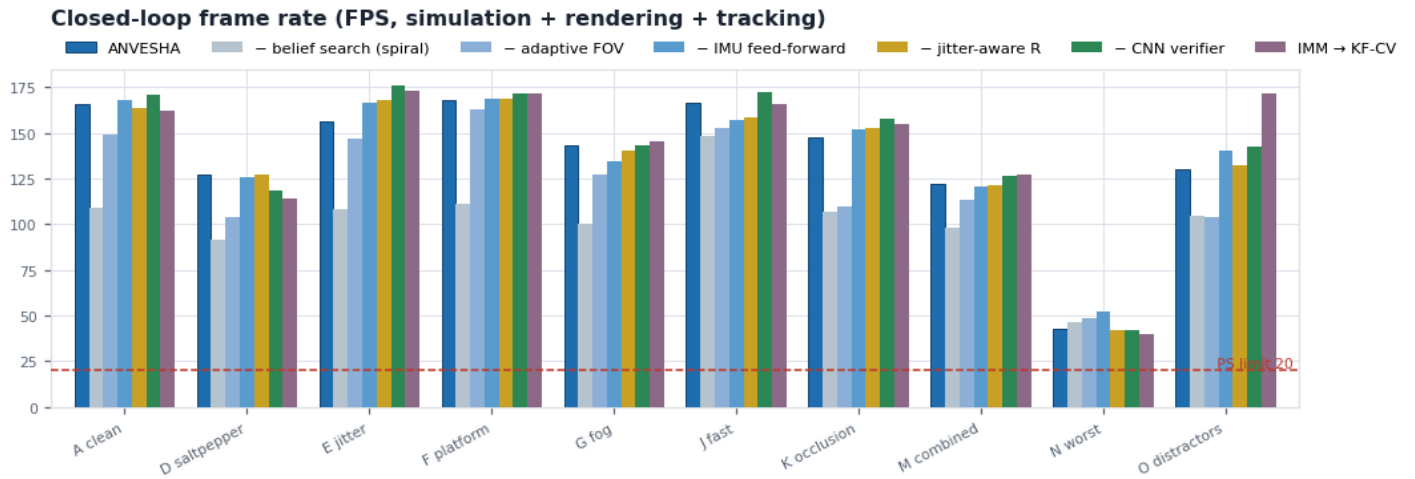
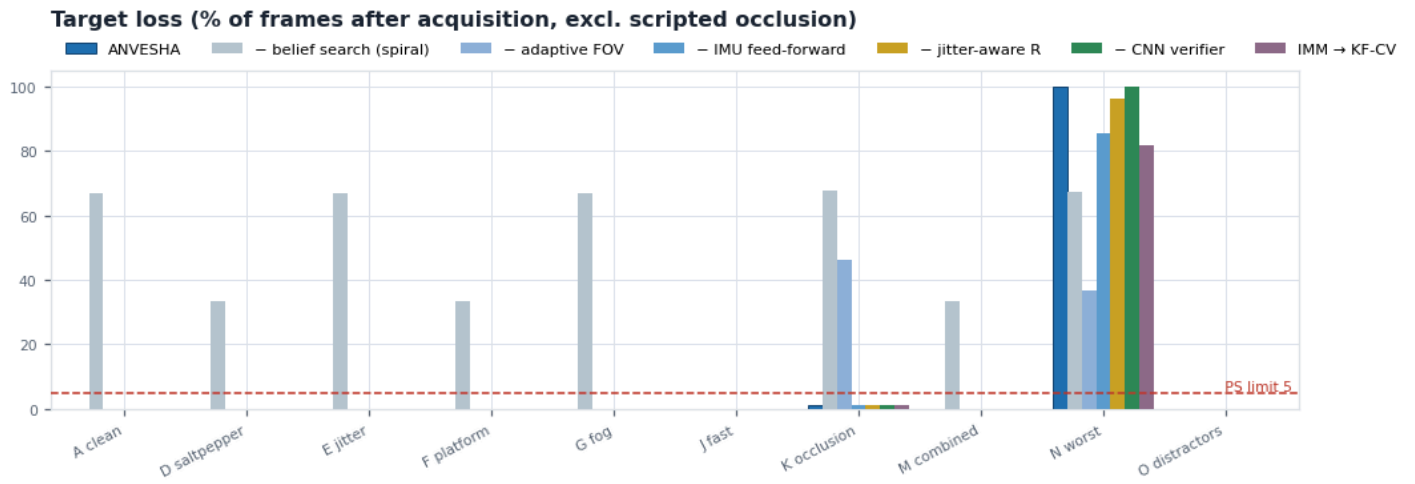
✓ = every limit met in the seed-aggregated result. Chips: **Acq** acquisition ≤ 2 s in every seed · **Err** mean tracking error ≤ 10 px (LOS and image) · **Loss** target loss $< 5\%$ (excl. scripted occlusion) · **Reacq** re-acquisition ≤ 1 s, none unrecovered · **FPS** closed-loop ≥ 20 FPS. Grey chip = not applicable (no loss event).

[illegible]

3 • Pipeline comparison

Bars are means over seeds. Red dashed line = PS limit. × = metric undefined because the pipeline never acquired; ▲ = value above the axis cap.





4 • Methodology & metric definitions

Simulation

240 Hz physics, 30 Hz 640×480 camera (4°×3° base FOV), 60 Hz control, pan/tilt limited to 5 °/s; seeded PCG64 streams per subsystem (bit-reproducible).

Truth isolation

The tracker receives only frames, encoder angles, IMU rates and timestamps; ground truth is used only by the metrics engine.

Baselines

A threshold+CoG+PID, B blob+KF+PID, C CNN+KF+PID; all use spiral search; PID gains grid-tuned (best set kp 12, ki 20, kd 0.05).

Acquisition time

Run start → first frame in TRACK with true LOS error ≤ 10 px. "Acq ≤ 2 s %" counts seeds meeting this.

LOS / image error

Distance between beacon and boresight after acquisition; LOS excludes the white camera-jitter component (uncorrectable by a coarse gimbal), image error includes it.

Target loss

% of frames after acquisition with LOS error > 10 px or no lock, excluding scripted occlusion windows.

Re-acquisition

Per loss episode, from loss (or end of scripted occlusion) to the next on-target frame; unrecovered episodes counted separately.

Centroiding error

Raw per-frame centroid vs true sub-pixel beacon position (camera px).

FPS

Closed-loop rate including scene rendering, tracking and control on the machine above.

5 • Full results

Green = PS limit met, red = not met. Means over seeds.

Scenario	Pipeline	Acq ≤2 s %	Acq mean s	Acq max s	LOS err px	Img err px	Centroid RMS px	Loss %	Re-acq max s	Unrecov.	False-lock fr	FPS
A clean	ANVESHA	100	1.29	1.60	1.48	1.48	0.010	0.0	—	0	0.0	166
A clean	— belief search (spiral)	0	7.83	7.83	1.47	1.47	0.008	66.7	—	0	0.0	109
A clean	— adaptive FOV	0	3.58	4.10	1.51	1.51	0.040	0.0	—	0	0.0	149
A clean	— IMU feed-forward	100	1.29	1.60	1.44	1.44	0.010	0.0	—	0	0.0	168
A clean	— jitter-aware R	100	1.29	1.60	1.48	1.48	0.010	0.0	—	0	0.0	164
A clean	— CNN verifier	100	1.29	1.60	1.48	1.48	0.010	0.0	—	0	0.0	171
A clean	IMM → KF-CV	100	1.29	1.60	2.03	2.03	0.010	0.0	—	0	0.0	163
D saltpepper	ANVESHA	100	1.12	1.37	6.74	6.74	0.106	0.0	—	0	0.0	127
D saltpepper	— belief search (spiral)	33	4.75	8.70	6.92	6.92	0.136	33.3	—	0	0.0	91
D saltpepper	— adaptive FOV	33	2.67	3.63	6.66	6.66	0.103	0.0	—	0	0.0	104
D saltpepper	— IMU feed-forward	100	1.12	1.37	6.74	6.74	0.106	0.0	—	0	0.0	126
D saltpepper	— jitter-aware R	100	1.12	1.37	6.69	6.69	0.107	0.0	—	0	0.0	127
D saltpepper	— CNN verifier	100	1.12	1.37	6.74	6.74	0.106	0.0	—	0	0.0	118
D saltpepper	IMM → KF-CV	100	1.14	1.40	8.69	8.69	0.109	0.0	—	0	0.0	114
E jitter	ANVESHA	100	1.34	1.60	4.63	8.89	0.010	0.0	—	0	0.0	157
E jitter	— belief search (spiral)	0	7.87	7.87	4.36	8.54	0.008	66.7	—	0	0.0	108
E jitter	— adaptive FOV	0	3.54	4.13	4.60	8.81	0.010	0.0	—	0	0.0	147
E jitter	— IMU feed-forward	100	1.36	1.60	4.60	8.87	0.010	0.0	—	0	0.0	167
E jitter	— jitter-aware R	67	2.69	4.90	19.26	20.50	0.010	0.0	—	0	0.0	168
E jitter	— CNN verifier	100	1.34	1.60	4.63	8.89	0.010	0.0	—	0	0.0	176
E jitter	IMM → KF-CV	100	1.33	1.57	5.77	9.42	0.010	0.0	—	0	0.0	173
F platform	ANVESHA	100	1.14	1.30	4.42	4.42	0.010	0.0	—	0	0.0	168
F platform	— belief search (spiral)	0	15.20	16.43	4.61	4.61	0.009	33.3	—	0	0.0	111
F platform	— adaptive FOV	67	1.61	2.90	4.49	4.49	0.028	0.0	—	0	0.0	163
F platform	— IMU feed-forward	100	1.16	1.27	5.76	5.76	0.010	0.0	—	0	0.0	168
F platform	— jitter-aware R	100	1.14	1.30	4.42	4.42	0.010	0.0	—	0	0.0	168
F platform	— CNN verifier	100	1.14	1.30	4.42	4.42	0.010	0.0	—	0	0.0	171
F platform	IMM → KF-CV	100	1.14	1.30	4.94	4.94	0.010	0.0	—	0	0.0	172
G fog	ANVESHA	100	1.29	1.60	1.48	1.48	0.091	0.0	—	0	0.0	143
G fog	— belief search (spiral)	0	7.83	7.83	1.47	1.47	0.063	66.7	—	0	0.0	100
G fog	— adaptive FOV	0	3.57	4.10	1.51	1.51	0.081	0.0	—	0	0.0	127
G fog	— IMU feed-forward	100	1.29	1.60	1.45	1.45	0.090	0.0	—	0	0.0	134
G fog	— jitter-aware R	100	1.29	1.60	1.48	1.48	0.091	0.0	—	0	0.0	140
G fog	— CNN verifier	100	1.29	1.60	1.48	1.48	0.091	0.0	—	0	0.0	143
G fog	IMM → KF-CV	100	1.29	1.60	2.03	2.03	0.091	0.0	—	0	0.0	145
J fast	ANVESHA	100	1.27	1.53	18.60	18.60	0.010	0.0	—	0	0.0	167
J fast	— belief search (spiral)	33	5.88	8.57	19.07	19.07	0.043	0.0	—	0	0.0	148
J fast	— adaptive FOV	33	2.72	3.73	18.61	18.61	0.008	0.0	—	0	0.0	153
J fast	— IMU feed-forward	100	1.27	1.53	18.72	18.72	0.010	0.0	—	0	0.0	157
J fast	— jitter-aware R	100	1.27	1.53	18.15	18.15	0.010	0.0	—	0	0.0	159
J fast	— CNN verifier	100	1.27	1.53	18.60	18.60	0.010	0.0	—	0	0.0	172
J fast	IMM → KF-CV	100	1.27	1.53	19.46	19.46	0.010	0.0	—	0	0.0	166
K occlusion	ANVESHA	100	1.29	1.60	16.18	16.18	0.013	1.2	0.27	0	0.0	148
K occlusion	— belief search (spiral)	0	8.07	8.07	5.72	5.72	0.009	67.7	0.33	0	0.0	106
K occlusion	— adaptive FOV	0	3.58	4.10	11.60	11.60	0.052	46.2	5.50	0	0.0	110
K occlusion	— IMU feed-forward	100	1.29	1.60	16.13	16.13	0.014	1.2	0.27	0	0.0	152
K occlusion	— jitter-aware R	100	1.29	1.60	16.18	16.18	0.013	1.2	0.27	0	0.0	153
K occlusion	— CNN verifier	100	1.29	1.60	16.18	16.18	0.013	1.2	0.27	0	0.0	158

K occlusion	IMM → KF-CV	100	1.29	1.60	16.94	16.94	0.014	1.2	0.27	0	0.0	155
M combined	ANVESHA	100	1.23	1.50	4.89	7.82	0.095	0.0	—	0	0.0	122
M combined	— belief search (spiral)	0	5.70	7.77	4.77	7.76	0.109	33.3	—	0	0.0	98
M combined	— adaptive FOV	67	2.74	6.23	4.98	7.88	0.091	0.0	—	0	0.0	113
M combined	— IMU feed-forward	100	1.27	1.57	6.98	9.04	0.097	0.0	—	0	0.0	120
M combined	— jitter-aware R	67	1.54	2.27	18.35	19.43	0.095	0.0	—	0	0.3	122
M combined	— CNN verifier	100	1.23	1.50	4.89	7.82	0.095	0.0	—	0	0.0	126
M combined	IMM → KF-CV	100	1.23	1.50	6.87	9.11	0.096	0.0	—	0	0.0	127
N worst	ANVESHA	0	—	—	—	—	0.551	100.0	—	0	150.3	43
N worst	— belief search (spiral)	0	7.23	7.23	34.66	37.96	0.564	67.3	0.07	0	61.3	46
N worst	— adaptive FOV	0	5.53	8.73	36.39	38.98	1.002	36.6	1.10	0	91.0	48
N worst	— IMU feed-forward	33	1.30	1.30	52.25	55.65	3.519	85.4	4.07	1	454.7	52
N worst	— jitter-aware R	0	16.80	16.80	38.75	44.58	16.685	96.2	0.07	1	267.7	42
N worst	— CNN verifier	0	—	—	—	—	0.613	100.0	—	0	174.7	42
N worst	IMM → KF-CV	0	6.67	6.67	36.62	38.48	1.343	81.8	3.83	0	109.0	40
O distractors	ANVESHA	67	2.29	3.70	1.50	1.50	0.018	0.0	—	0	23.0	130
O distractors	— belief search (spiral)	0	7.64	10.60	1.52	1.52	0.031	0.0	—	0	44.0	105
O distractors	— adaptive FOV	0	8.04	12.77	1.62	1.62	0.064	0.0	—	0	58.7	104
O distractors	— IMU feed-forward	67	2.33	3.70	1.46	1.46	0.018	0.0	—	0	22.7	140
O distractors	— jitter-aware R	67	2.29	3.70	1.50	1.50	0.018	0.0	—	0	23.0	132
O distractors	— CNN verifier	33	3.43	4.60	1.48	1.48	0.019	0.0	—	0	44.0	143
O distractors	IMM → KF-CV	67	2.29	3.70	2.05	2.05	0.018	0.0	—	0	23.0	171

6 • Limitations & honest notes

- All results are from simulation; no real camera or gimbal was used. The learned verifier is trained and evaluated on synthetic patches only.
- Uniform ± 20 px white jitter alone gives ≈ 16.3 px radial RMS image error — above 10 px for any coarse gimbal; a fine-steering stage is required.
- Vibration above the gimbal bandwidth and very fast manoeuvres raise LOS error above 10 px; scripted occlusions add error while the beacon is invisible.
- The all-maxima stress case (every PS disturbance at its maximum plus fog and look-alikes) is not solved.